

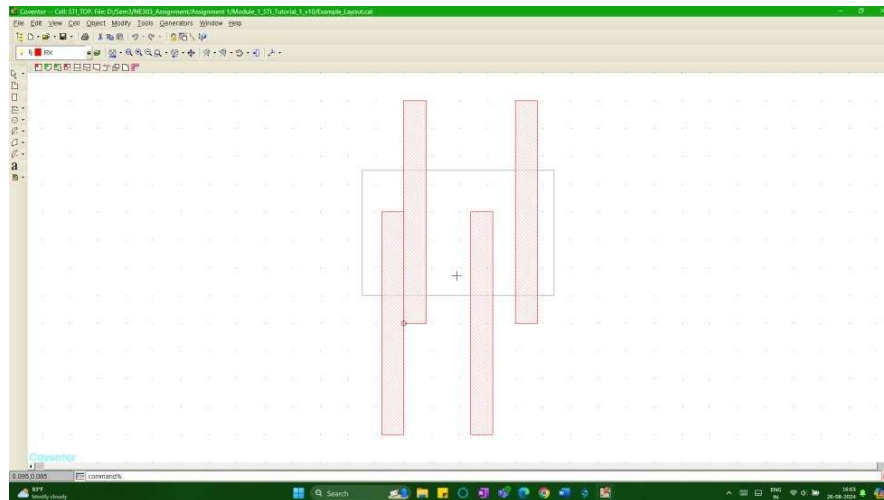
Assignment 1: Installation

Module_1_STI_Tutorial_1_v10

Table below shows the key steps in the STI process flow.

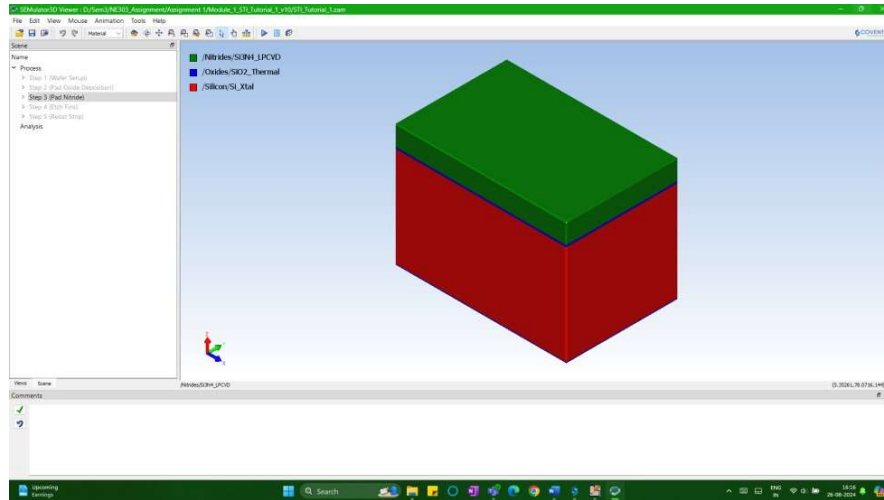
Part	Key Learning	Task
0	Open the Deck	Open tutorial and view its components
1	Planar Deposit	Spin on Photoresist
	Expose Material	Photolithography
2	Etch	Etch Trench
3	Deposit	Deposit Trench Oxide
	Polish	Polish Trench Oxide, stopping on Nitride
4	3D Viewer	Export Image and Animation

First, we start with a mask as shown below, the outer boundary is defining the size of wafer in 2D and rest are phot mask used for photolithography in software.

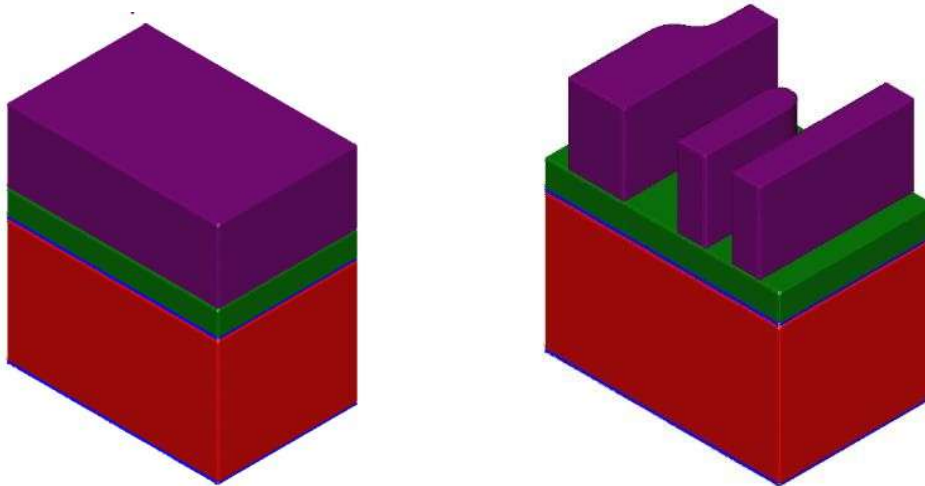


In the next image we are seeing process steps to be done for fabricating our device.

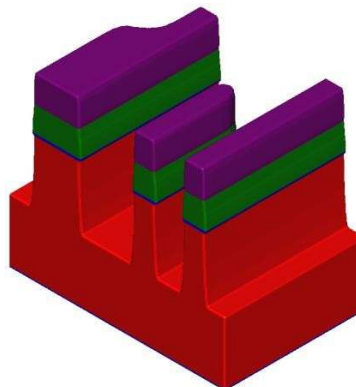
When the simulation is finished, layout viewer is automatically opened and shows the device structure as shown below.



After adding Lithography (Resist coating and exposing) step, we get a device structure as shown below.

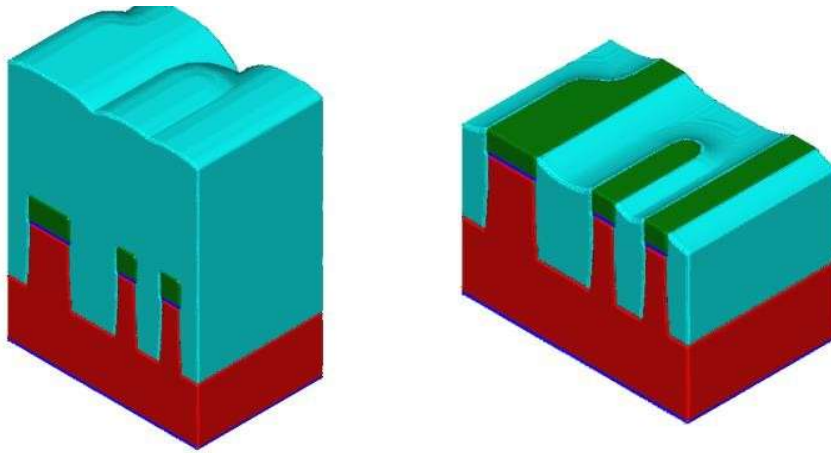


Adding etch steps gives a high aspect ratio fin as shown below,

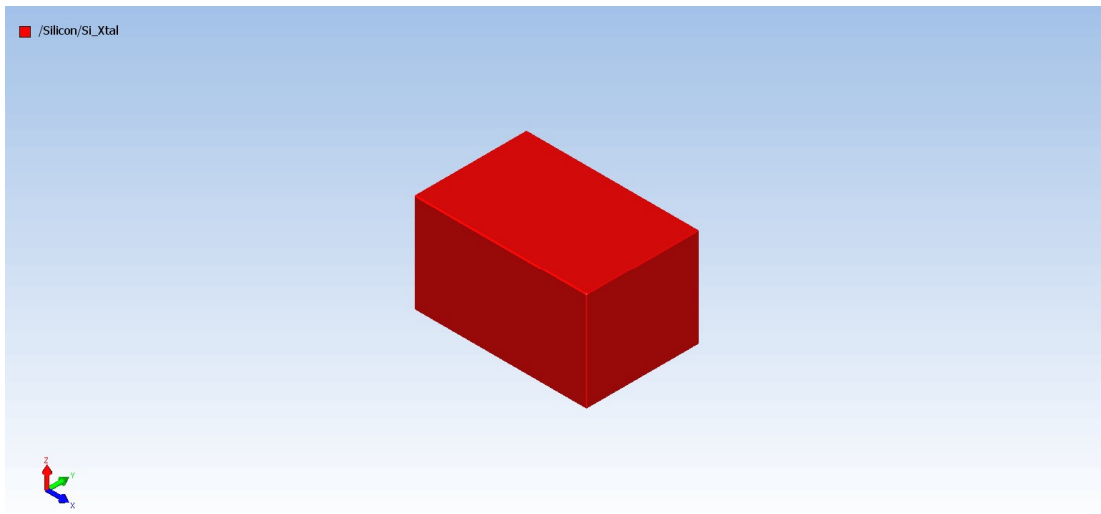


After depositing STI we get device structure

oxide and polishing it as below,



Using export option, I have created an animation file as below.



Module_2_AdvTraining_Tutorial_Start_10

Table below shows the key steps in the BEOL process flow.

Part	Key Learning	Task
1	Multi Etch	BEOL trench overetch with chamfering
2	Visibility Deposit	Non-conformal deposition of liner films
3	Virtual Metrology	Measure via bottom CD, via bottom contact area
4	Structure Search	Count number of electrical nets; locate regions with min line width and thin dielectric
5	Expeditor	Conduct 2-level, 2-factor DOE on M4/V3

Image below shows Layout of the mask layer wise,

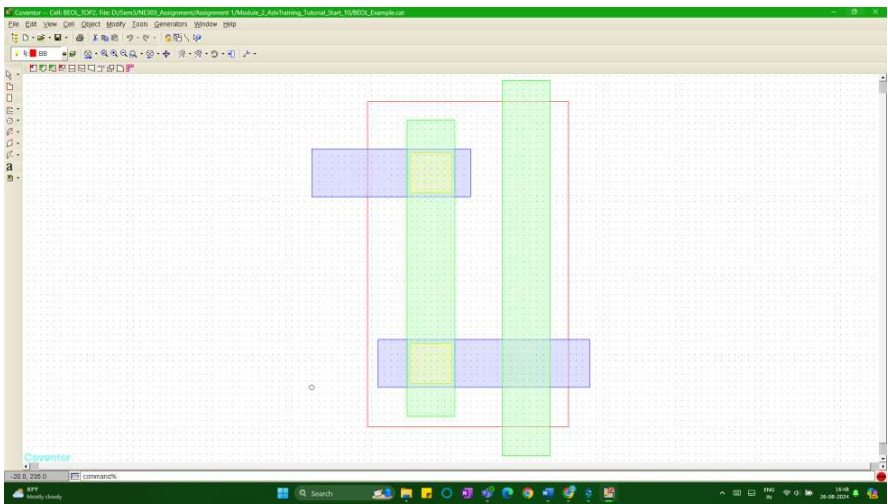
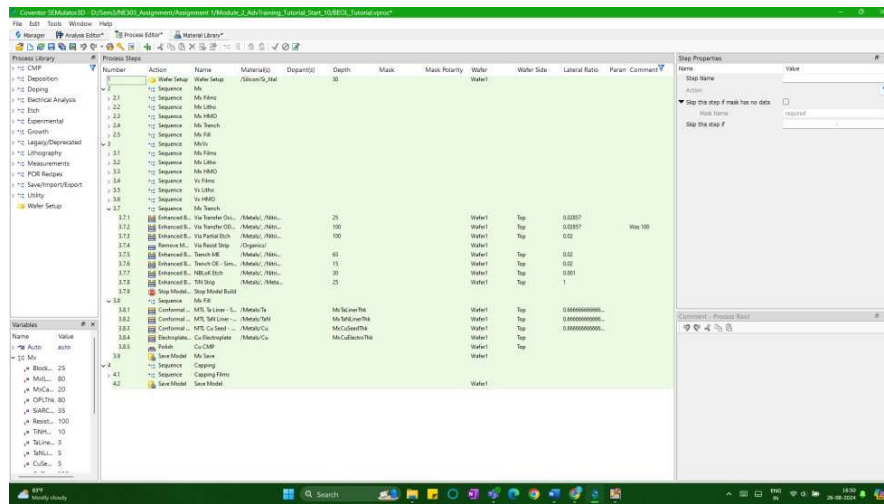
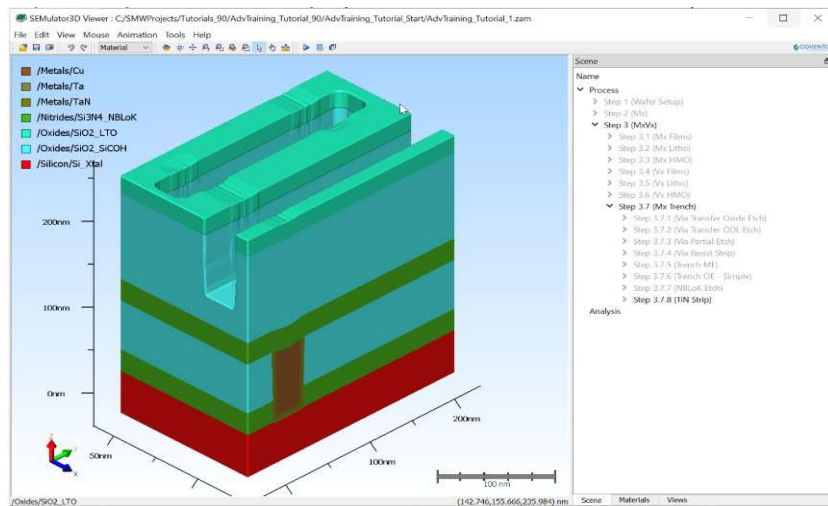


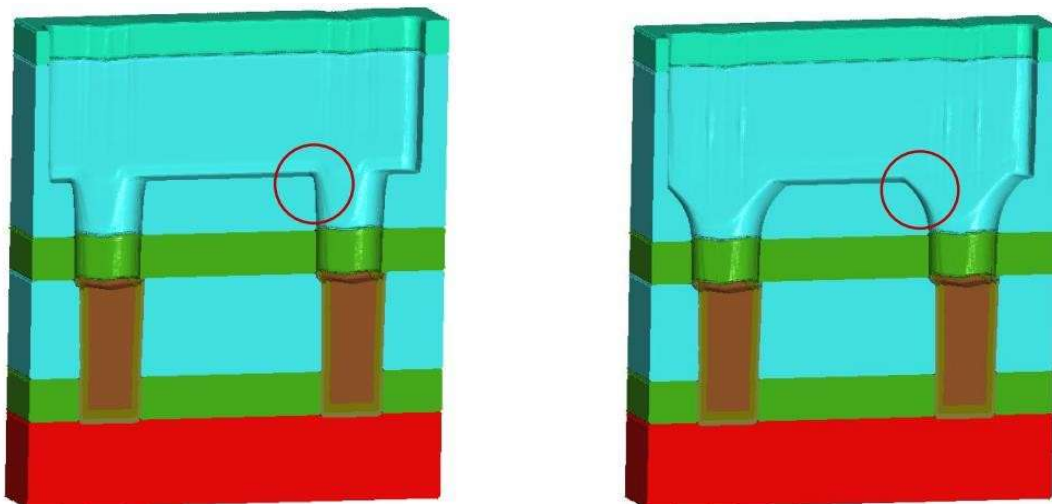
Image below shows process flow,



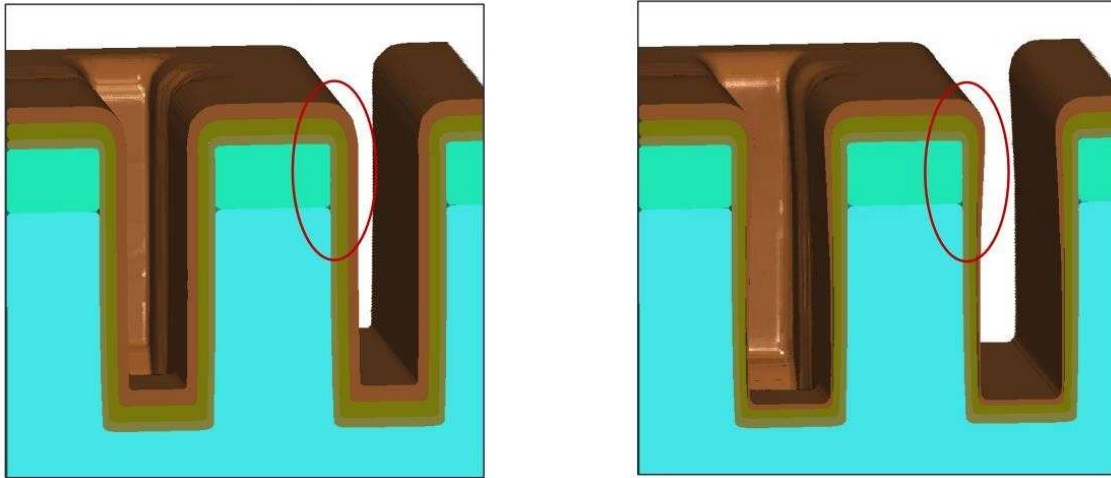
Initially model builds look like below,



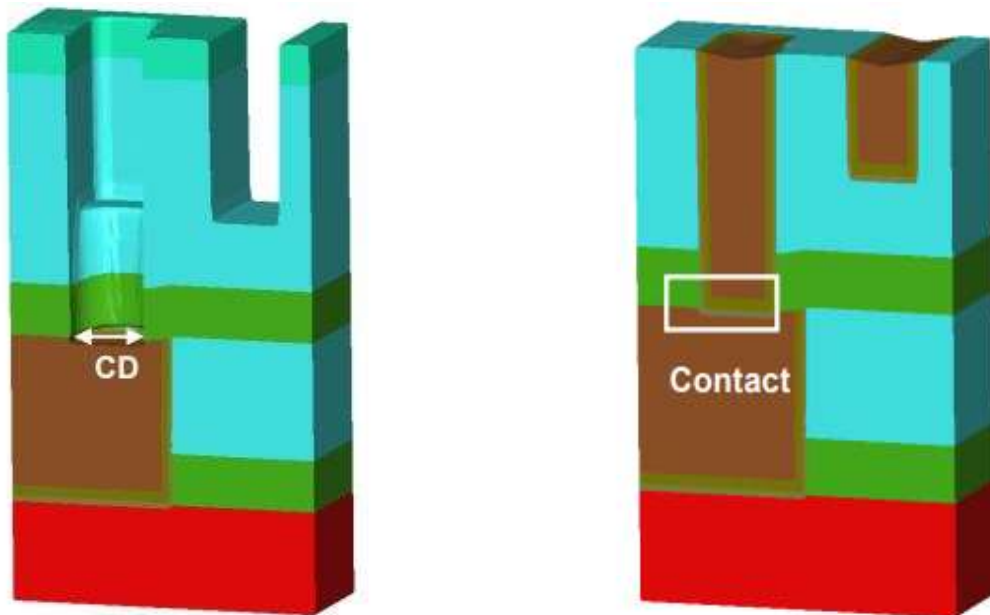
After changing basic trench etch to multi-etch the corners are slightly over etched, as shown by comparison (left- Basic Etch, right- Multi Etch) below,



After changing deposition technique from conformal to visibility the side walls are deposited with less material than on corners, as shown by comparison (left- Conformal deposition, right- Visibility deposition) below,

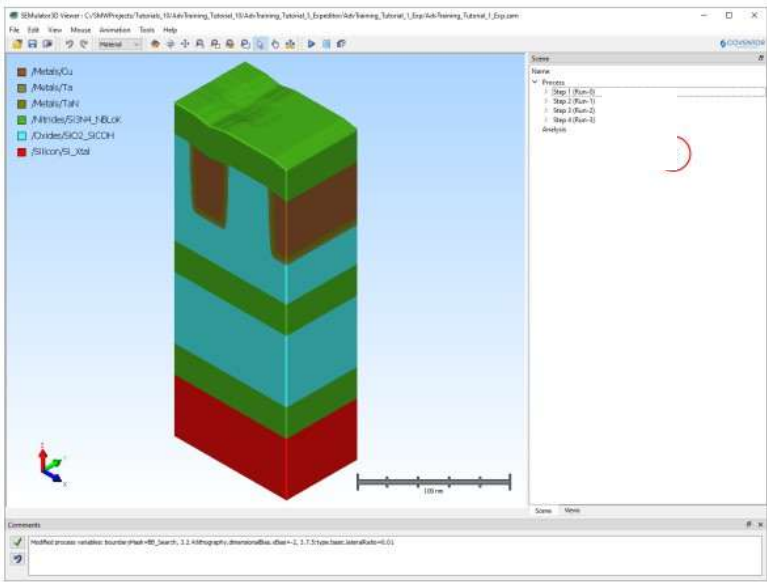


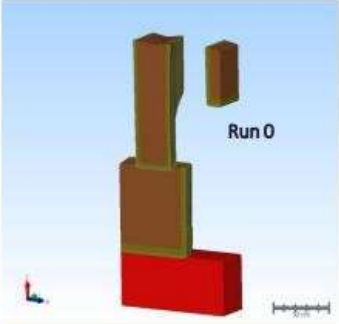
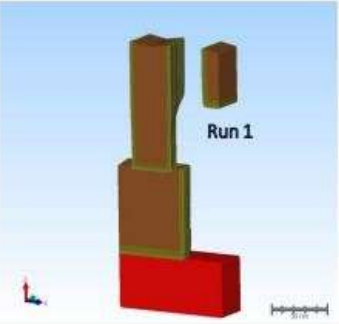
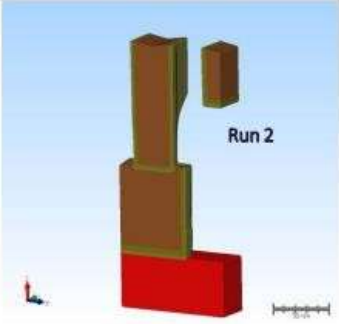
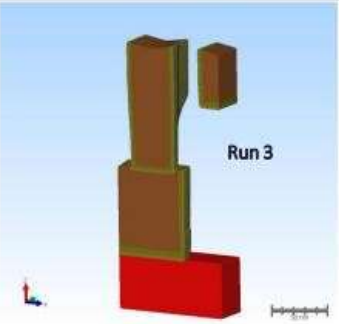
Using virtual metrology, we measured CD and Contact area (left and right below) of via,



Measured CD is 39.826 nm and contact area is 1278.480 nm².

Following image and table shows expeditor results,



		Trench ME Basic Etch Lateral Ratio	
		0.01	0.05
Litho Mx Dimensional xBias	-2	 Run 0	 Run 1
	2	 Run 2	 Run 3